

Authentication & Protected Routes (JWT)

Task Title

Authentication & Protected Routes using JWT (Node.js + Express + React + MongoDB)

Objective

The goal of this task is to implement a secure authentication system using **JWT (JSON Web Token)**. It includes user signup, login, password hashing, token generation, and protecting routes on both backend and frontend.

Tools & Technologies Used

- Node.js
- Express.js
- MongoDB
- Mongoose
- React.js
- JWT (jsonwebtoken)
- bcryptjs
- Postman

Features Implemented

User Authentication

- User Signup (Registration)
- User Login
- Password hashing using bcrypt

JWT Authentication

- JWT token generated on login
- Token used for route protection
- Token verification middleware implemented

Protected Routes

- Only authenticated users can access protected APIs
- Unauthorized users are blocked

Frontend Features

- Login form created in React
- Token stored securely (recommended: httpOnly cookie)
- Protected routes implemented in React (route guarding)
- Redirect to login if not authenticated

API Flow

1. User registers → password is hashed using bcrypt
2. User logs in → credentials verified
3. JWT token generated and sent to client
4. Token stored securely (cookie/local storage)
5. Token sent with API requests
6. Backend verifies token → allows access to protected routes

Security Implementation

Password Hashing

- Passwords are hashed using **bcryptjs** before storing in database
- Prevents storing plain text passwords

JWT Authentication

- Tokens are signed using secret key
- Tokens expire after a set time

Protected Routes

- Middleware verifies token before allowing access
-

Where to Store Tokens

Recommended (Best Practice)

- **httpOnly cookies**
 - Prevents XSS attacks
 - More secure for production apps

Alternative (Less Secure)

- localStorage
 - Easy to use
 - Vulnerable to XSS attacks

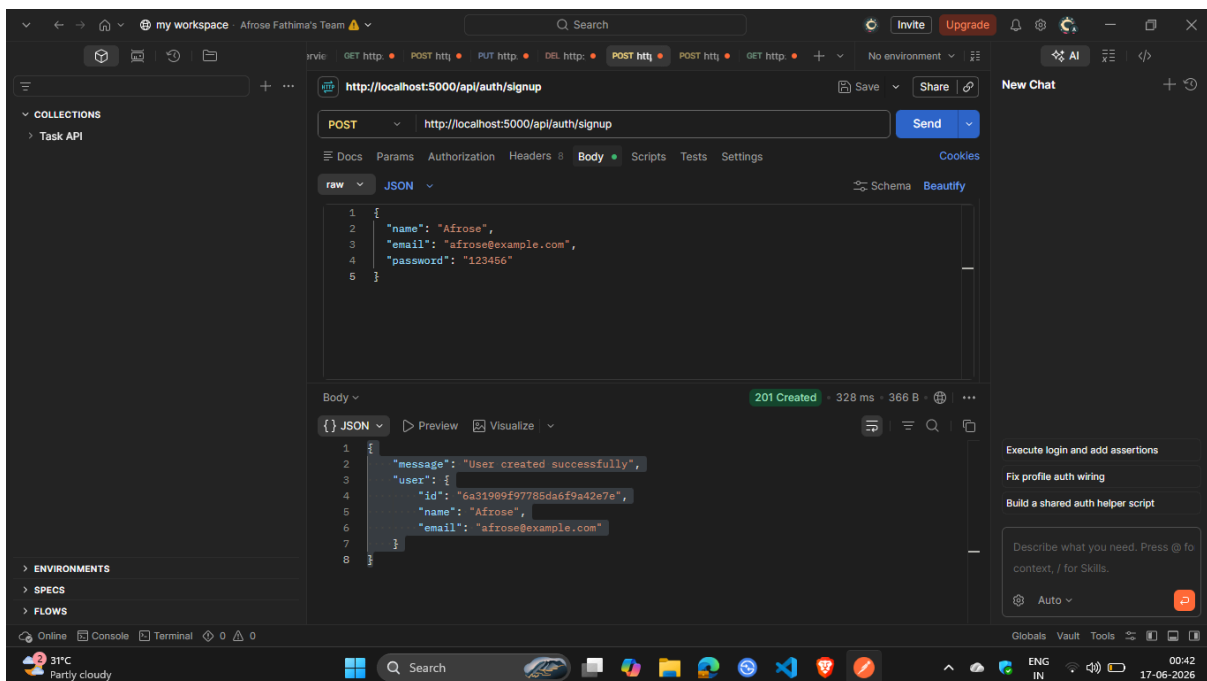
Common Mistakes / Pitfalls

- Storing passwords without hashing
- Exposing JWT secret key
- Not validating tokens in backend
- Storing tokens in insecure storage
- Forgetting token expiry setup

What I Learned

- How authentication works in modern web apps
- How to secure passwords using hashing
- How JWT enables stateless authentication
- How to protect frontend and backend routes
- How to build secure login system

Screenshots



my workspace Afrose Fathima's Team

Search

Menu

COLLECTIONS

- Task API

ENVIRONMENTS

SPECS

FLOWS

Online Console Terminal

31°C Partly cloudy

http://localhost:5000/api/auth/login

POST http://localhost:5000/api/auth/login

Send

Docs Params Authorization Headers Body Scripts Tests Settings Cookies

raw JSON

```
1 {
2   "email": "afrose@example.com",
3   "password": "123456"
4 }
5
```

Body

200 OK · 197 ms · 424 B

JSON

```
1 {
2   "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ3a2V5SWQ1IiYTMkOTA5ZjJkN2ZlZGE2Zj1hMDJlN2U1LCJpYXQiOiJlZ300E2MzNjYsImV4cCI6MTc4MTcxOTc2Nm8uCu1v3akYvIglQ9IXF9_XzuyIuzAMcFwyDgDsdEfbiY"
3 }
```

Execute login and add assertions

Fix profile auth wiring

Build a shared auth helper script

Describe what you need. Press @ for context, / for Skills.

Auto

Globals Vault Tools

ENG IN 00:43 17-06-2026

my workspace Afrose Fathima's Team

Search

Menu

COLLECTIONS

- Task API

ENVIRONMENTS

SPECS

FLOWS

Online Console Terminal

31°C Partly cloudy

http://localhost:5000/profile

GET http://localhost:5000/profile

Send

Docs Params Authorization Headers Body Scripts Tests Settings Cookies

Headers

Key	Value	Description	Bulk Edit	Presets	...
Authorization	Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ3a2V5SWQ1IiYTMkOTA5ZjJkN2ZlZGE2Zj1hMDJlN2U1LCJpYXQiOiJlZ300E2MzNjYsImV4cCI6MTcxOTc2Nm8uCu1v3akYvIglQ9IXF9_XzuyIuzAMcFwyDgDsdEfbiY				
Key	Value	Description			

Body

200 OK · 18 ms · 366 B

JSON

```
1 {
2   "message": "Protected route accessed successfully",
3   "user": {
4     "userId": "6a31989f97785da6f9a42e7e",
5     "iat": 1781633366,
6     "exp": 1781719766
7   }
8 }
```

Execute login and add assertions

Fix profile auth wiring

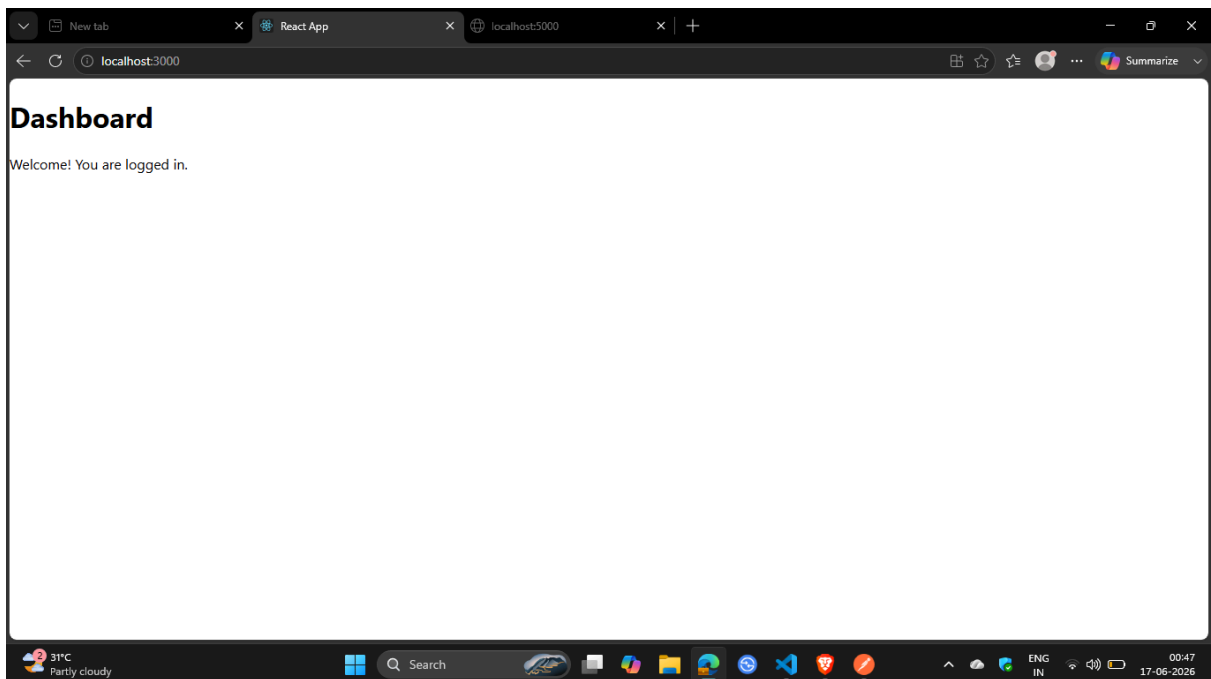
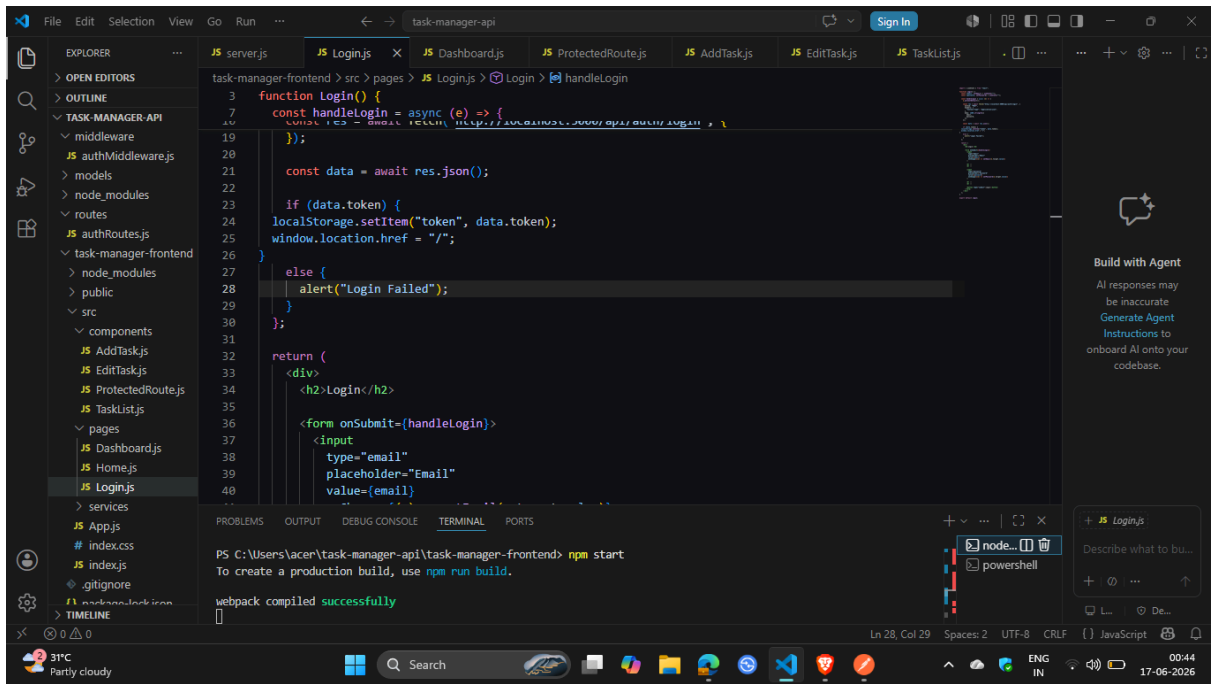
Build a shared auth helper script

Describe what you need. Press @ for context, / for Skills.

Auto

Globals Vault Tools

ENG IN 00:43 17-06-2026



GitHub Repository

<https://github.com/afrosefathima455-hub/task-manager-api.git>